Speed summary, 2026-08-29. Every timing comparison on disk, per lane, per vendor

Compiled 2026-08-29 from the boards already under `bench/results/`. Nothing here was rerun. Every number carries the file it came from and the commit or date that file records. Where a file and a newer file disagree on one cell, both are shown and the newer one is marked current.

1. How to read this

Three vendor columns. Apple M4 (the MacBook Pro on the desk, 10 cores, 16 GB), NVIDIA H100 80GB SXM (RunPod), AMD Instinct MI325X (DigitalOcean `tor1`). The peer on NVIDIA and AMD is the competitor's own GPU arm and nothing else (cuML, cuVS, CatBoost-GPU, XGBoost-GPU, LightGBM-CUDA, cuBLAS, hipBLASLt, torch on CUDA or ROCm). The harness refuses the CPU arms on those boxes by name (`GPU-PATH-ONLY` in every refusal table). On Apple the peer is the CPU arm of the same library, because CatBoost's `task_type="GPU"` raises on Apple silicon, LightGBM ships no Metal learner, and cuML and cuVS are CUDA-only (`bench/results/fast_speed/2026-08-26-APPLE-trees.md`). The one exception the boards record is the gemm and sequence lanes, where torch on MPS is a real Apple GPU peer and is used as such (`bench/results/fast_speed/2026-08-28-APPLE-gemmseq.md`). Every AMD row is our arm alone, because no peer library runs on the MI325X (`bench/results/BOARD_2026-08-28_three-vendor.md` section 2.3).

Two tiers. FAST is the default build and promises speed only. IDENTICAL is the `-D MOJOLEARN_NUMERIC_IDENTICAL=1` build and promises the same bits on three vendors. Every `fast_speed/` board is FAST on both sides; every leg records `ours_headers_identical=0`. The IDENTICAL timings live in `bench/results/SPEED_LANE_2026-08-25.md` and `bench/results/lanes_price/`, and are labeled IDENTICAL wherever they appear below. Boards older than 2026-08-23 predate the tier split and do not record a mode; they are labeled "mode not recorded". A row never mixes tiers without saying so.

`ratio` is median(ours) over median(theirs) as the boards compute it, restated here as "Nx faster" or "Nx slower". Medians are over the timed rounds after one untimed warm-up, ours and the peer alternating inside one process. On the M4 that alternation is the only defense against the thermal drift that file after file records (section 5).

2. Per family, per vendor, from the boards

Column key. ours ms and peer ms are medians. tier is the mode of OUR arm. src is the board and the commit or date it records.

2.1 Gradient boosting, symmetric trees (CatBoost pair only)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1,000,000 x 28, 5 rounds	856.116	catboost-gpu 864.690	1.01x faster	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-forest-A.md`, a8d838e, 2026-08-28 (current)
higgs 2,000,000 x 28, 5 rounds	1827.444	catboost-gpu 1379.246	1.32x slower	FAST	NVIDIA H100	same
higgs 1,000,000 x 28, 5 rounds	3666.967 (min 2622.429)	catboost-gpu 846.102	4.33x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28_030908-nvidia-forest.md`, 3d65d70, 2026-08-28 earlier leg; superseded by forest-A above, see section 5
higgs 2,000,000 x 28, 5 rounds	1304.338	catboost-gpu 1227.208	1.06x slower	FAST	NVIDIA H100	same
higgs 5,000,000 x 28, 5 rounds	5227.542	catboost-gpu 2458.494	2.13x slower	FAST	NVIDIA H100	same (the only 5M symmetric row at a 2026-08-28 commit)
higgs 1M / 2M / 5M, 3 rounds	668.142 / 1185.877 / 2760.228	catboost-gpu 994.561 / 1312.121 / 2543.586	1.49x faster / 1.11x faster / 1.09x slower	FAST	NVIDIA H100	`fast_speed/2026-08-26-nvidia-forest-postfix.md`, 132d754, 2026-08-26 (older)
higgs 1M / 5M, 3 rounds	636.590 / 2998.741	catboost-gpu 1182.268 / 2687.290	1.86x faster / 1.12x slower	FAST	NVIDIA H100	`fast_speed/2026-08-26_040049-nvidia-trees-ladder.md`, c637e30, 2026-08-26 (older)
year 463,715 x 90, 3 rounds	1931.296	catboost-cpu 2197.476	1.14x faster	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-forest.md`, a8d838e, 2026-08-28 (current)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
year 463,715 x 90, 3 rounds	2005.700	catboost-cpu 2411.320	1.20x faster	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea, 2026-08-27 (older)
higgs 1,000,000 x 28, 3 rounds	4409.991	catboost-cpu 5762.869	1.31x faster	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea, 2026-08-27 (current for this cell)
higgs 1,000,000 x 28, 3 rounds	3762.358	catboost-cpu 3806.136	1.01x faster, parity	FAST	Apple M4	`fast_speed/2026-08-26-APPLE-HIGGS-1M.md`, 5b728e2, 2026-08-26 (older)
year, gbm-bench 500 rounds depth 8	11.95 s	cat-cpu 13.59 s	1.14x faster	mode not recorded	Apple M4	`THREE_SUITE_QUIET_2026-08-22.md`, dfa41bb, 2026-08-22; same row in `README.md` benchmark table
higgs 8.8M, gbm-bench 500 rounds	131.5 s	cat-cpu 177.8 s	1.35x faster, AUC 0.822 vs 0.830 open gap	mode not recorded	Apple M4	same
synthclf 720,000 x 100, 10 rounds	1041.722	none	no peer on AMD	FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest.md`, 88b918d, 2026-08-28; synthetic fallback fixture, see section 3
higgs 1M / 2M	unrun	unrun		FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest-higgs.md`, 4f6a17a: our arm raised at warm-up (DEVIATION 1910, 64-lane half-byte histograms)

2.2 Gradient boosting, depthwise

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1M x 28, 5 rounds	1382.644	catboost-gpu 1139.637	1.21x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-forest-A.md`, a8d838e (current)
higgs 1M x 28, 5 rounds	1382.644	xgboost-gpu 743.058	1.86x slower	FAST	NVIDIA H100	same
higgs 2M x 28, 5 rounds	2483.338	catboost-gpu 1496.789	1.66x slower	FAST	NVIDIA H100	same
higgs 2M x 28, 5 rounds	2483.338	xgboost-gpu 1310.138	1.90x slower	FAST	NVIDIA H100	same
higgs 5M x 28, 5 rounds	5287.681	catboost-gpu 2559.735 / xgboost-gpu 2165.906	2.07x / 2.44x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28_030908-nvidia-forest.md`, 3d65d70 (only 5M depthwise row)
higgs 1M / 2M, 3 rounds	1316.996 / 3112.093	xgboost-gpu 685.180 / 1141.118	1.92x / 2.73x slower	FAST	NVIDIA H100	`fast_speed/2026-08-27_171939-nvidia-depthwise.md`, bc2a3b4, 2026-08-27 (older)
year 463,715 x 90, 3 rounds	4543.493	catboost-cpu 4189.724	1.08x slower	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-forest.md`, a8d838e (current)
year 463,715 x 90, 3 rounds	6224.937	catboost-cpu 4880.259	1.28x slower	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea (older)
synthclf 720,000 x 100, 10 rounds	1446.487	none	no peer on AMD	FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest.md`, 88b918d
higgs 1M / 2M	unrun			FAST	AMD MI325X	our arm raised at warm-up, DEVIATION 1910

2.3 Gradient boosting, lossguide (leaf-wise)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1M x 28, 5 rounds	1751.063	catboost-gpu 1533.086 / xgboost-gpu 771.236	1.14x / 2.27x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-forest-B.md`, a8d838e (current)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 2M x 28, 5 rounds	2690.113	catboost-gpu 1864.952 / xgboost-gpu 1185.849	1.44x / 2.27x slower	FAST	NVIDIA H100	same
higgs 1M x 28, 5 rounds	1906.192	lightgbm-cuda 1313.705	1.45x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28_030911-nvidia-forest.md`, 3d65d70 (the only leg where lightgbm-cuda ran)
higgs 2M x 28, 5 rounds	2936.414	lightgbm-cuda 1669.264	1.76x slower	FAST	NVIDIA H100	same
higgs 5M x 28, 5 rounds	6301.633	catboost-gpu 2870.462 / xgboost-gpu 2413.373	2.20x / 2.61x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28_030908-nvidia-forest.md`, 3d65d70 (current 5M row)
higgs 5M x 28, 3 rounds	14800.158 (min 6030.534)	xgboost-gpu 2647.444	5.59x slower	FAST	NVIDIA H100	`fast_speed/2026-08-27_031904-nvidia-forest.md`, 69f2503, 2026-08-27 (older; the 5M rung, section 5)
year 463,715 x 90, 3 rounds	8660.290	catboost-cpu 5894.057 / lightgbm-cpu 3886.938	1.47x / 2.23x slower	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-forest.md`, a8d838e (current)
year 463,715 x 90, 3 rounds	10822.723	catboost-cpu 6267.296	1.73x slower	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea (older)
synthclf 720,000 x 100, 10 rounds	2437.570	none	no peer on AMD	FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest.md`, 88b918d
higgs 1M / 2M	unrun			FAST	AMD MI325X	our arm raised at warm-up, DEVIATION 1910

2.4 Random forest

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1M x 28, 5 rounds	5309.306	cuml-rf-gpu 3177.072	1.67x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-forest-A.md`, a8d838e (current)
higgs 2M x 28, 5 rounds	7308.012	cuml-rf-gpu 4445.280	1.64x slower	FAST	NVIDIA H100	same
higgs 5M x 28, 5 rounds	12224.530	cuml-rf-gpu 7284.710	1.68x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28_030908-nvidia-forest.md`, 3d65d70
higgs 1M / 2M / 5M, 3 rounds	6323.739 / 8286.882 / 14047.177	cuml-rf-gpu 3486.530 / 4839.604 / 7598.814	1.81x / 1.71x / 1.85x slower	FAST	NVIDIA H100	`fast_speed/2026-08-27_031904-nvidia-forest.md`, 69f2503 (older)
higgs 1M / 5M, 3 rounds	6188.417 / 14698.891	cuml-rf-gpu 3079.464 / 7057.536	2.01x / 2.08x slower	FAST	NVIDIA H100	`fast_speed/2026-08-26_040049-nvidia-trees-ladder.md`, c637e30 (older)
covtype 522,911 x 54, 3 rounds	15281.972	sklearn-rf-cpu 12430.019	1.23x slower	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-forest.md`, a8d838e (current)
covtype 522,911 x 54	15281.972	lightgbm-cpu 258148.588 (n=1, budget exceeded)	16.89x faster	FAST	Apple M4	same
covtype 522,911 x 54, 3 rounds	19875.178	sklearn-rf-cpu 13721.545	1.45x slower	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea (older)
higgs 1M x 28, 3 rounds	26774.701	sklearn-rf-cpu 82120.529	3.07x faster	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea (current for this cell)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1M x 28, 3 rounds	26909.764	sklearn-rf-cpu 84216.725	3.13x faster	FAST	Apple M4	`fast_speed/2026-08-26-APPLE-HIGGS-1M.md`, 5b728e2 (older)
covtype, gbm-bench 100 trees	4.02 s	sklearn RF 10 cores 5.26 s	1.31x faster	mode not recorded	Apple M4	`THREE_SUITE_QUIET_2026-08-22.md`, dfa41bb; `README.md` table reads 4.0 s vs 5.3 s, 1.33x
year, gbm-bench 100 trees	14.14 s	sklearn RF 10 cores 385.1 s	27.2x faster (sleep-straddled invocation, flagged)	mode not recorded	Apple M4	same
higgs 8.8M, gbm-bench 100 trees	47.05 s	sklearn RF 10 cores 310.3 s	6.6x faster	mode not recorded	Apple M4	same; `README.md` 47.1 s vs 310.3 s
covtype, gbm-bench pairs	4.62 s	lgbm-rf-cpu 12.70 s	2.75x faster	mode not recorded	Apple M4	`RF_ET_2026-08-22_lightgbm-pairs.md`, 2026-08-22
year, gbm-bench pairs	23.09 s	lgbm-rf-cpu 7.69 s	3.0x slower	mode not recorded	Apple M4	same
higgs 8.8M, gbm-bench pairs	68.12 s	lgbm-rf-cpu 29.53 s	2.3x slower, ours wins every accuracy column	mode not recorded	Apple M4	same
higgs 1M x 28, 5 rounds	4525.381	none	no peer on AMD; H100 same source 5309.306, AMD 1.17x faster than H100	FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest-higgs.md`, 4f6a17a; cross-box row in `BOARD_2026-08-28_three-vendor.md` 2.3
higgs 2M x 28, 5 rounds	6268.397	none	H100 7308.012, AMD 1.17x faster	FAST	AMD MI325X	same
synthclf 720,000 x 100, 10 rounds	3057.926	none		FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest.md`, 88b918d

2.5 Extra trees

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1M x 28	4935.292 (5 rounds)	lightgbm-cuda 181070.935 (n=1, budget exceeded)	36.69x faster, and better logloss 0.622 vs 0.644	FAST	NVIDIA H100	`fast_speed/2026-08-28_030911-nvidia-forest.md`, 3d65d70
higgs 2M x 28	7449.510 (5 rounds)	lightgbm-cuda 227402.966 (n=1)	30.53x faster	FAST	NVIDIA H100	same
higgs 1M / 2M, 5 rounds	4160.298 / 7661.135	none ran (cuML has no ExtraTrees; lightgbm-cuda build refused in this leg)	no peer	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-forest-B.md`, a8d838e
higgs 5M x 28, 5 rounds	14826.190	none ran	no peer	FAST	NVIDIA H100	`fast_speed/2026-08-28_030908-nvidia-forest.md`, 3d65d70
covtype 522,911 x 54, 3 rounds	9826.842	sklearn-et-cpu 11006.799	1.12x faster, acc 0.6796 vs 0.6768	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-forest.md`, a8d838e (current)
covtype 522,911 x 54	9826.842	lightgbm-cpu 6625.966	1.48x slower, lgbm acc 0.5417	FAST	Apple M4	same
covtype 522,911 x 54, 3 rounds	14648.080	sklearn-et-cpu 12972.721	1.13x slower	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea (older)
covtype 522,911 x 54, 3 rounds	9843.1	sklearn-et-cpu 10454.7	1.06x faster	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-covtype-et-flip.md`, f732d11 (older)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
higgs 1M x 28, 3 rounds	23754.352	sklearn-et-cpu 29630.341	1.25x faster	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-trees-evening.md`, 20729ea (current for this cell)
higgs 1M x 28, 3 rounds	20944.393	sklearn-et-cpu 28761.257	1.37x faster	FAST	Apple M4	`fast_speed/2026-08-26-APPLE-HIGGS-1M.md`, 5b728e2 (older)
covtype, gbm-bench 100 trees	3.70 s	sklearn ET 10 cores 5.44 s	1.47x faster	mode not recorded	Apple M4	`THREE_SUITE_QUIET_2026-08-22.md`, dfa41bb; `README.md` 3.7 s vs 5.4 s, 1.46x
year, gbm-bench 100 trees	18.84 s	sklearn ET 10 cores 54.64 s	2.90x faster (flagged invocation)	mode not recorded	Apple M4	same
higgs 8.8M, gbm-bench 100 trees	39.02 s	sklearn ET 10 cores 132.4 s	3.39x faster	mode not recorded	Apple M4	same; `README.md` 39.0 s vs 132.4 s, 3.4x
covtype, gbm-bench pairs	4.10 s	lgbm-et-cpu 6.84 s	1.67x faster	mode not recorded	Apple M4	`RF_ET_2026-08-22_lightgbm-pairs.md`
year, gbm-bench pairs	15.60 s	lgbm-et-cpu 6.17 s	2.5x slower, MSE 98.0 vs 92.8 ours behind	mode not recorded	Apple M4	same
higgs 1M x 28, 5 rounds	18294.084	none	H100 same source 4160.298, AMD 4.40x slower	FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest-higgs.md`, 4f6a17a
higgs 2M x 28, 5 rounds	35521.428	none	H100 7661.135, AMD 4.64x slower	FAST	AMD MI325X	same
synthclf 720,000 x 100, 10 rounds	24199.518	none		FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest.md`, 88b918d

2.6 Isolation forest

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
anomaly 500,000 x 32, 10 rounds	232.422	cuml-iforest-gpu 85.159	2.73x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-forest-B.md`, a8d838e
anomaly 500,000 x 32, 3 rounds	219.910	sklearn-iforest-cpu 147.265	1.49x slower	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-forest.md`, a8d838e
anomaly 500,000 x 32, 10 rounds	155.077	none	H100 232.422, AMD 1.50x faster than H100	FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest-higgs.md`, 4f6a17a
anomaly 500,000 x 32, 10 rounds	153.199	none		FAST	AMD MI325X	`fast_speed/2026-08-28-AMD-forest.md`, 88b918d

2.7 k-means

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
4,000,000 x 32, k=64, 20 iter, 5 rounds	155.353	cuml-gpu 120.217	1.29x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91, 2026-08-28 (current)
same, 5 rounds	155.067	cuml-gpu 236.789	1.53x faster	FAST	NVIDIA H100	`fast_speed/2026-08-26_040100-nvidia-classical.md`, 2026-08-26 (older; the cuML arm moved 236.8 to 120.2 ms between legs)
same, 5 rounds	167.145	cuml-gpu 233.434	1.40x faster	FAST	NVIDIA H100	`fast_speed/2026-08-25_235543-nvidia-classical.md`, 2026-08-25 (older)
4,000,000 x 32, k=64, 20 iter, 3 rounds	748.774	none ran (cuML not on this box)	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
4M x 32, k=64, 20 iter	759.49	sklearn KMeans 2055.22	2.71x faster, ranges disjoint	mode not recorded	Apple M4	`BOARD_2026-08-20_algorithm-matched.md`, 2026-08-20
any	unrun				AMD MI325X	no AMD classical FAST speed leg exists under `fast_speed/`

2.8 k-NN (brute force) and IVF

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
knn 400,000 x 32, 4,000 q, k=10, 5 rounds	26.404	cuml-gpu 9.555	2.76x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91 (current; DEV 1921-1923 landed)
same	233.827 / 243.862	cuml-gpu 10.822 / 10.224	21.61x / 23.85x slower	FAST	NVIDIA H100	`fast_speed/2026-08-26_040100-nvidia-classical.md` and `2026-08-25_235543-nvidia-classical.md` (older; quoted as the target row in `LANE_knn-speed-campaign_2026-08-28.md`)
knn own arms, H100	ours 26.404, ours-fused 26.409, ours-tiled 53.926		AUTO matches fused	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`
knn 400,000 x 32, 4,000 q, k=10, 3 rounds	692.947	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
knn own arms, Apple	ours 692.947, ours-fused 699.921, ours-tiled 616.570 (hash moved 3 of 3 rounds)		tiled 1.12x faster than shipped choice	FAST	Apple M4	same
knn 400k idx, 4k q, d=32, k=10	695.94	sklearn 943.53	1.36x faster	mode not recorded	Apple M4	`BOARD_2026-08-20_algorithm-matched.md`, 2026-08-20
ivf 512x8, q64, L8, p3, k8, 5 rounds	7.143	cuvs-gpu 14.584	2.04x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
ivf same, 3 rounds	86.763	none ran (cupy missing)	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`
knn or ivf	unrun				AMD MI325X	no AMD classical FAST speed leg

2.9 DBSCAN and HDBSCAN

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
dbscan 4,000 x 16, 5 rounds	0.682	cuml-gpu 1.093	1.60x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
hdbscan blobs96 96 x 4, 5 rounds	1.162	cuml-gpu 4.417	3.80x faster (fixed-cost row, no size knob)	FAST	NVIDIA H100	same
dbscan 4,000 x 16, 3 rounds	22.431	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
hdbscan 96 x 4, 3 rounds	21.551	none ran	no peer	FAST	Apple M4	same
dbscan 4k x 16	11.75	sklearn 9.08	0.77x, INDISTINGUISHABLE (ranges overlap)	mode not recorded	Apple M4	`BOARD_2026-08-20_algorithm-matched.md`

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
dbscan d=8, 50k to 800k, n_jobs=-1	172.9 to 7930.5	sklearn 93.1 to 4009.6	0.43x to 0.66x, a loss above ~20k points	mode not recorded	Apple M4	`DBSCAN_RBC_2026-08-19.md`, 2026-08-19 late
dbscan or hdbscan	unrun				AMD MI325X	no AMD classical FAST speed leg

2.10 PCA and OLS

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
pca 4,000,000 x 32, 8 comp, 5 rounds	9.278	cuml-gpu (jacobi) 119.669	12.90x faster	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
ols 4,000,000 x 32, 5 rounds	8.555	cuml-gpu (eig) 118.837	13.89x faster	FAST	NVIDIA H100	same
ols 4,000,000 x 32, 5 rounds	8.607	cuml-gpu 176.938	20.56x faster	FAST	NVIDIA H100	`fast_speed/2026-08-26_040100-nvidia-classical.md` (older; pca refused there, `svd_solver` name)
pca 4M x 32 c8, 3 rounds	36.315	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
ols 4M x 32, 3 rounds	33.500	none ran	no peer	FAST	Apple M4	same
pca 4M x 32, 8 comp	37.63	sklearn covariance_eigh 132.53	3.52x faster	mode not recorded	Apple M4	`BOARD_2026-08-20_algorithm-matched.md`, 2026-08-20
ols 4M x 32	34.79	sklearn Ridge cholesky 149.22	4.29x faster	mode not recorded	Apple M4	same
pca or ols	unrun				AMD MI325X	no AMD classical FAST speed leg

sklearn's own GPU path on Apple was measured and is slower than its CPU path (PCA full on MPS 1484.6 ms against covariance_eigh CPU 136.6 ms; `SKLEARN_GPU_BASELINE_2026-08-20.md`), so the CPU arm is their best arm on that box.

2.11 Ridge, logistic, lasso and elasticnet (cd), cholesky, krr

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
cd 2048 x 16, 5 rounds	1.137	cuml-gpu 1.283	1.13x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
cholesky RBF 64x64 r4, 5 rounds	0.341	torch-gpu (cuSOLVER) 0.162	2.11x slower (fixed-cost row)	FAST	NVIDIA H100	same
krr RBF 16x5, 5 rounds	0.300	cuml-gpu 1.825	6.08x faster (fixed-cost row)	FAST	NVIDIA H100	same
cd 2048 x 16, 3 rounds	7.630	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
cholesky, 3 rounds	4.737	none ran (torch not installed in that env)	no peer	FAST	Apple M4	same
krr, 3 rounds	4.962	none ran	no peer	FAST	Apple M4	same
ridge, logistic	unrun on any speed board				all	no `ridge` or `logreg` lane exists in the classical speed harness; they have identity cards only (`CHANGELOG.md` 0.2.0)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
cd, cholesky, krr	unrun				AMD MI325X	no AMD classical FAST speed leg

2.12 SVM

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
F2 xor 240 x 2, fit only, 5 rounds	1.333	cuml-gpu 4.864	3.65x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
same, 3 rounds	12.044	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
svm	unrun				AMD MI325X	no AMD classical FAST speed leg

2.13 KDE

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
1024 x 256 x 8, 5 rounds	0.203	cuml-gpu 0.385	1.89x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
same, 3 rounds	2.691	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
kde	unrun				AMD MI325X	no AMD classical FAST speed leg

2.14 Linkage (single linkage hierarchy)

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
blobs_dups 102 x 5, 5 rounds	0.963	cuml-gpu 2.318	2.41x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
same, 3 rounds	14.035	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
linkage	unrun				AMD MI325X	no AMD classical FAST speed leg

2.15 Spectral, GMM, GP, Nystroem, RBFSampler, resample

RAPIDS ships none of these, so on NVIDIA and AMD they have no legal peer. On Apple the peer is scikit-learn or SciPy on the CPU, and the boards mark every one of these rows FIXED-COST at tiny shipped fixtures (24x2 to 48x4), "NOT a speed claim".

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
spectral 48 x 4 c3, 3 rounds	75.866	sklearn-cpu 3.648	20.80x slower, fixed cost	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7 (current)
spectral, 6 rounds	84.643	sklearn-cpu 3.180	26.62x slower	FAST	Apple M4	`fast_speed/2026-08-27-APPLE-classical-six-small-lanes.md`, 9a4e759 (older)
gmm SEPARATED 24 x 2 K3, 3 rounds	32.283	sklearn-cpu 1.115	28.95x slower, fixed cost	FAST	Apple M4	`2026-08-28-APPLE-classical.md` (current); 35.52x on 2026-08-27
gp ard 12 x 3 s6, 3 rounds	6.125	sklearn-cpu 0.290 (float64, no optimizer both sides)	21.14x slower, fixed cost	FAST	Apple M4	same (current); 24.78x on 2026-08-27
nystroem RBF 16 x 5 q8, 3 rounds	5.816	sklearn-cpu 0.483	12.03x slower, fixed cost	FAST	Apple M4	same (current); 13.87x on 2026-08-27
rbfsampler 3 x 5 q8, 3 rounds	2.580	sklearn-cpu 0.167	15.41x slower, fixed cost	FAST	Apple M4	same (current); 19.04x on 2026-08-27

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
resample 200 x 2, r10000, 3 rounds	7.673	scipy-cpu 35.149	4.58x faster	FAST	Apple M4	same (current); 4.01x on 2026-08-27
spectral, gmm, gp, nystroem, rbfsampler, resample	ours only, 2.456 / 0.464 / 0.418 / 0.193 / 0.529 / 6.428	none	no peer, RAPIDS ships none	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`
all six	unrun				AMD MI325X	no AMD classical FAST speed leg

2.16 Time series (holtwinters, kpss); ARIMA

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
holtwinters 7 x 72 f12, 5 rounds	10.399	cuml-gpu 8.156	1.27x slower (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
kpss 8 x 520, 5 rounds	0.130	cuml-gpu 0.436	3.35x faster (fixed-cost row)	FAST	NVIDIA H100	same
holtwinters, 3 rounds	17.816	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
kpss, 3 rounds	0.681	none ran (statsmodels missing)	no peer	FAST	Apple M4	same
holtwinters, kpss	unrun				AMD MI325X	no AMD classical FAST speed leg
arima	unrun on every column				all	`CHANGELOG.md` 0.2.0, `ARIMA` has no `fit`; `estimate_x0` and the batched L-BFGS driver are not ported

2.17 Metrics

dataset and shape	ours ms	peer and ms	ratio	tier	vendor	src
eleven metrics, lab2053 flt2053 sil521x4 tru301x6, 5 rounds	0.687	cuml-gpu 12.626	18.37x faster (fixed-cost row)	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-classical.md`, 65d5e91
same, 3 rounds	9.491	none ran	no peer	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-classical.md`, cc499f7
metrics	unrun				AMD MI325X	no AMD classical FAST speed leg

2.18 GEMM and matmul

FAST rows first. DEVIATION 1885 records that the FAST vendor GEMM on NVIDIA is a TF32-class kernel (MAX `linalg.matmul` measured 200 TFLOP/s beside cuBLAS TF32 207.5 and cuBLAS FP32 44.4), so on NVIDIA the fair FAST peer is `cublas-tf32` (`fast_speed/2026-08-26-DEVIATION-1885-fast-gemm-is-tf32.md`, `SPEED_LANE_2026-08-25.md` 1.1).

shape	ours ms	peer and ms	ratio	tier	vendor	src
gram.128sq.x100003, 5 rounds	0.142	cublas-fp32 0.093 / cublas-tf32 0.058	1.53x / 2.47x slower	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-gemmseq.md`, a8d838e
gram.32x32x1M	0.271	cublas-fp32 0.237 / tf32 0.108	1.14x / 2.50x slower	FAST	NVIDIA H100	same
gram.32x32x64K	0.047	0.037 / 0.027	1.27x / 1.74x slower	FAST	NVIDIA H100	same
kmeans.dist.4096x64x64	0.025	0.020 / 0.017	1.25x / 1.45x slower	FAST	NVIDIA H100	same

shape	ours ms	peer and ms	ratio	tier	vendor	src
ols.step1.16x16x64K	0.045	0.036 / 0.027	1.27x / 1.65x slower	FAST	NVIDIA H100	same
ols.predict.gemv.64Kx16	0.016	0.018 / 0.017	1.12x / 1.10x faster	FAST	NVIDIA H100	same
pca.transform.8192x4x4	0.021	0.019 / 0.018	1.11x / 1.16x slower	FAST	NVIDIA H100	same
pca.transform.wide.8192x64x128	0.029	0.023 / 0.019	1.28x / 1.51x slower	FAST	NVIDIA H100	same
llama8b.qkv t1 / t8 / t512	0.031 / 0.054 / 0.090	tf32 0.040 / 0.044 / 0.066	1.29x faster / 1.22x slower / 1.37x slower	FAST	NVIDIA H100	same
llama8b.mlp_up t1 / t8 / t512	0.083 / 0.115 / 0.211	tf32 0.093 / 0.105 / 0.185	1.11x faster / 1.10x slower / 1.14x slower	FAST	NVIDIA H100	same
llama8b.mlp_down t1 / t8 / t512	0.085 / 0.110 / 0.232	tf32 0.094 / 0.103 / 0.211	1.11x faster / 1.07x slower / 1.10x slower	FAST	NVIDIA H100	same
llama8b.lm_head t8 / t512	0.775 / 1.407	tf32 0.751 / 1.357	1.03x / 1.04x slower	FAST	NVIDIA H100	same; lm_head.t1 ours refused (m=1 gemv path, `SPEED_LANE` section 4)
gram.32x32x1M, 3 rounds	8.041	mps-default 8.340	1.04x faster	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-gemmseq.md`, 8fd467f
gram.128sq.x100003	4.212	mps-default 1.894	2.22x slower	FAST	Apple M4	same
gram.32x32x64K	0.856	0.756	1.13x slower	FAST	Apple M4	same
kmeans.dist.4096x64x64	0.279	0.260	1.07x slower	FAST	Apple M4	same
ols.step1.16x16x64K	0.705	0.761	1.08x faster	FAST	Apple M4	same
ols.predict.gemv.64Kx16	0.503	0.239	2.11x slower	FAST	Apple M4	same
pca.transform.8192x4x4	0.209	0.220	1.05x faster	FAST	Apple M4	same
pca.transform.wide.8192x64x128	0.431	0.329	1.31x slower	FAST	Apple M4	same
llama8b.qkv t1 / t8 / t512	0.947 / 2.067 / 17.047	mps-default 0.990 / 1.195 / 6.508	1.05x faster / 1.73x slower / 2.62x slower	FAST	Apple M4	same
llama8b.mlp_up t1 / t8 / t512	3.052 / 9.404 / 36.506	2.976 / 3.892 / 22.494	1.03x / 2.42x / 1.62x slower	FAST	Apple M4	same
llama8b.mlp_down t1 / t8 / t512	2.997 / 4.853 / 53.140	2.763 / 3.539 / 22.665	1.08x / 1.37x / 2.34x slower	FAST	Apple M4	same
llama8b.lm_head t1 / t8 / t512	24.148 / 40.071 / 251.424	22.179 / 29.002 / 238.973	1.09x / 1.38x / 1.05x slower	FAST	Apple M4	same
any FAST gemm shape	unrun			FAST	AMD MI325X	no AMD gemmseq FAST speed leg under `fast_speed/`; the only AMD gemm timings are IDENTICAL, below

IDENTICAL rows, strict FP32 on our side, from `SPEED_LANE_2026-08-25.md` (commits ba35096 through f5b905a, 2026-08-25). Read that file's section 1 first; the M4 and MI325X columns are single-round and the H100 column is a band across two single-round legs.

shape	ours	peer	ratio	tier	vendor	src
llama8b.qkv.t1	2.21 ms	MAX linalg.matmul 1.21 / MPS 1.41	1.8x / 1.6x slower	IDENTICAL vs vendor FAST	Apple M4	`SPEED_LANE_2026-08-25.md` 3.1

shape	ours	peer	ratio	tier	vendor	src
llama8b.qkv.t8	4.50	1.41 / 2.72	3.2x / 1.7x slower	IDENTICAL	Apple M4	same
llama8b.qkv.t512	218.8	11.6 / 8.7	18.9x / 25.1x slower	IDENTICAL	Apple M4	same
llama8b.mlp_up.t512	771.5	39.8 / 25.8	19.4x / 29.9x slower	IDENTICAL	Apple M4	same
llama8b.mlp_down.t512	770.4	68.9 / 29.0	11.2x / 26.6x slower	IDENTICAL	Apple M4	same
llama8b.lm_head.t8	113.1	39.5 / 32.3	2.9x / 3.5x slower	IDENTICAL	Apple M4	same
peak achieved	ours 78 GF/s	MPS 2332 GF/s	about 2 percent of what the box gives torch	IDENTICAL	Apple M4	same
llama8b.qkv.t512	1.95 to 3.51 TF/s	cuBLAS strict FP32 44.4 TF/s (TF32 207.5)	12.7x to 22.8x slower	IDENTICAL vs FP32	NVIDIA H100	`SPEED_LANE_2026-08-25.md` 3.2
llama8b.mlp_up.t512	2.00 to 3.55	42.8 (TF32 291.3)	12.1x to 21.4x slower	IDENTICAL	NVIDIA H100	same
llama8b.mlp_down.t512	1.96 to 3.47	49.8 (TF32 263.6)	14.4x to 25.4x slower	IDENTICAL	NVIDIA H100	same
llama8b.qkv.t512	4.31 TF/s	hipBLASLt strict FP32 77.1	17.9x slower	IDENTICAL	AMD MI325X	`SPEED_LANE_2026-08-25.md` 3.3 (torch 2.9.1+rocm6.4, 10 repeats, median)
llama8b.mlp_up.t512	4.52	70.9	15.7x slower	IDENTICAL	AMD MI325X	same
llama8b.mlp_down.t512	4.37	50.4	11.5x slower	IDENTICAL	AMD MI325X	same
llama8b.lm_head.t8	3.81	12.9	3.4x slower	IDENTICAL	AMD MI325X	same
share of FP32 vector peak	2 to 5 percent on all three vendors			IDENTICAL	all three	`SPEED_LANE_2026-08-25.md` 3.4

2.19 Sequence lanes (attention, mlp, rmsnorm, transformer, mamba)

These are FAST. On NVIDIA every lane routing a GEMM took the TF32 cut of DEVIATION 1885 and the file says to quote its speedups only with "part of this is a TF32 precision cut" attached. Best and worst peer per shape from the 2026-08-28 boards; full per-arm rows are in the source files.

lane and shape	ours ms	peer and ms	ratio	tier	vendor	src
attention lane.b2_l4_d32_kv2	0.284	torch-gpu-sdpa-efficient-fp32 0.295 to torch-gpu-fp32 0.398	1.04x to 1.40x faster	FAST	NVIDIA H100	`fast_speed/2026-08-28-NVIDIA-gemmseq.md`, a8d838e
attention llama8b.prefill.t512	2.171	torch-gpu-tf32 0.642 to torch-gpu-fp32 1.425	3.38x to 1.52x slower	FAST	NVIDIA H100	same
attention llama8b.prefill.t8	1.314	sdpa-efficient 0.311	4.23x slower (worst)	FAST	NVIDIA H100	same
mlp llama8b.prefill.t512	0.684	torch-gpu-fp32 4.046 / tf32 0.716	5.92x faster / 1.05x faster	FAST	NVIDIA H100	same
mlp llama8b.prefill.t8	0.338	fp32 0.513 / tf32 0.306	1.51x faster / 1.11x slower	FAST	NVIDIA H100	same
rmsnorm lane.b2_l4_d32_kv2	0.010	torch-gpu-fp32 0.053	5.40x faster	FAST	NVIDIA H100	same

lane and shape	ours ms	peer and ms	ratio	tier	vendor	src
rmsnorm llama8b.prefill.t128	1.052	tf32 0.062	16.88x slower (worst row on the board)	FAST	NVIDIA H100	same
transformer lane.b2_l4_d32_kv2	0.396	torch-gpu-fp32 0.540	1.36x faster	FAST	NVIDIA H100	same
transformer llama8b.prefill.t512	8.880	tf32 1.380 / fp32 5.542	6.44x / 1.60x slower	FAST	NVIDIA H100	same
transformer llama8b.decode.t1.ctx512	3.594	sdpa-math 0.623	5.77x slower	FAST	NVIDIA H100	same
mamba mamba130m.prefill.t512	2.470	torch-ref-scan-gpu 21.236	8.60x faster (peer is the pure-PyTorch reference scan; mamba_ssm not importable)	FAST	NVIDIA H100	same
mamba mamba130m.prefill.t1 / t8 / t128	0.237 / 0.350 / 1.042	torch-ref-scan-gpu 0.414 / 0.754 / 5.691	1.75x / 2.16x / 5.46x faster	FAST	NVIDIA H100	same
transformer llama8b.prefill.t1, before and after DEV 1873-1877	1009.778 then 1.885	best torch arm	686x slower became 7.12x slower at t512	FAST	NVIDIA H100	`fast_speed/2026-08-26-DEVIATION-1885-fast-gemm-is-tf32.md`, 2026-08-26
attention lane.b2_l4_d32_kv2	4.282	torch-gpu-fp32 2.302 to flash-bf16 0.557	1.86x to 7.68x slower	FAST	Apple M4	`fast_speed/2026-08-28-APPLE-gemmseq.md`, 8fd467f
attention llama8b.prefill.t512	88.574	fp32 27.636 to flash-bf16 18.524	3.21x to 4.78x slower	FAST	Apple M4	same
attention llama8b.prefill.t8	20.762	flash-bf16 1.565	13.27x slower (worst)	FAST	Apple M4	same
mlp llama8b.decode.t1.ctx512	7.744	fp32 7.374 / tf32 7.062	1.05x / 1.10x slower	FAST	Apple M4	same
mlp llama8b.prefill.t512	111.150	fp32 71.586	1.55x slower	FAST	Apple M4	same
rmsnorm lane.b2_l4_d32_kv2	0.269	fp32 0.494 / tf32 0.440	1.84x / 1.64x faster	FAST	Apple M4	same
rmsnorm llama8b.decode.t1.ctx512	3.277	tf32 0.226	14.48x slower (worst)	FAST	Apple M4	same
transformer llama8b.prefill.t512	221.114	fp32 101.787 / flash-bf16 80.126	2.17x / 2.76x slower	FAST	Apple M4	same
transformer lane.b2_l4_d32_kv2	8.707	flash-fp32 1.086	8.02x slower (worst)	FAST	Apple M4	same
mamba, all five shapes	4.480 to 7.749	none ran (einops missing, mamba_ssm missing)	no peer	FAST	Apple M4	same
every sequence lane	unrun			FAST	AMD MI325X	no AMD gemmseq FAST speed leg

Row counts as the boards state them. NVIDIA gemmseq 2026-08-28 has 53 rows with an opponent, all fixed-cost by the undeclared-scale rule; the three vendor board counts the same leg as 130 comparison pairs, 55 wins and 75 losses (`BOARD_2026-08-28_three-vendor.md` 2.1). Apple gemmseq has 44 rows with an opponent; the board calls it 116 pairs.

3. What is unrun, and why the boards say so

cell	why
AMD MI325X, gbdt symmetric, depthwise, lossguide on higgs 1M and 2M	Our arm raised at warm-up in the 4f6a17a leg ("half-byte histograms on a 64-lane column ... DEVIATION 1910"), `fast_speed/2026-08-28-AMD-forest-higgs.md`. FAST gbdt builds and runs on the MI325X since DEVIATIONS 1906 and 1910 (commits f853e8df and 19b319c7, `e1/2026-08-29_093711-mojolearn-e2-amd/p9_fast_build_gbdt.log`, `stability/` gbdt lanes STABLE 6/6) but the speed rows have not been rerun (`BOARD_2026-08-28_three-vendor.md` 2.3).

cell	why
AMD MI325X, any peer in any lane	No legal GPU opponent exists on AMD. cuBLAS, cuML, cuVS, CatBoost-GPU, XGBoost-GPU, LightGBM-CUDA are CUDA-only; each arm prints `FSPEED-REFUSED` (`BOARD_2026-08-28_three-vendor.md` 2.3). catboost-gpu on the MI325X died with "CUDA driver version is insufficient" and lightgbm-cuda with "CUDA Tree Learner was not enabled" (`2026-08-28-AMD-forest-higgs.md`).
AMD MI325X, all 21 classical lanes and all gemm and sequence lanes, FAST	No AMD classical or gemmseq FAST speed leg exists under `bench/results/fast_speed/`; the only AMD leg directories are `do-2026-08-28_*-amd-forest`.
AMD MI325X, gemm XF32 arm	`SPEED_LANE_2026-08-25.md` 3.3 states the harness does not touch `allow_tf32` on the hip backend, so the CDNA3 XF32 arm is owed the way TF32 is measured on NVIDIA.
AMD synthetic-fixture leg (88b918d)	The leg had no dataset download step, so every higgs rung refused and fell back to `synthclf-720000x100`; "60 REAL timed rounds, none on the dataset the comparison needs" (`BOARD_2026-08-28_three-vendor.md` defect 9). The 4f6a17a leg fixed it for rf, et and iforest.
NVIDIA H100, extra trees against cuML	cuML has no ExtraTrees estimator (refusal text in every NVIDIA forest board). The only GPU peer that ever ran is lightgbm-cuda, once, at 3d65d70, with n=1 because it exceeded the 300 s per-arm budget.
NVIDIA H100, iforest at higgs 1M or 2M	The iforest lane ships the anomaly 500k fixture only; no higgs rung exists for it on any board.
NVIDIA H100, gemm llama8b.lm_head.t1	Our FAST arm refused; MAX `linalg.matmul` crashes at m=1, n=128256 on an H100 (`SPEED_LANE_2026-08-25.md` section 4, DEVIATION 1093).
NVIDIA H100, llama8b.lm_head.t512 in the IDENTICAL lane	Above the 5e10 MAC cap on both sides, skipped on both (`SPEED_LANE_2026-08-25.md` section 2).
NVIDIA H100, torch flash arms for attention and transformer	`No available kernel` at every shape on that box (`2026-08-28-NVIDIA-gemmseq.md` refusals).
NVIDIA H100, mamba against mamba-ssm	`mamba_ssm` not importable on any box; the only peer is the pure-PyTorch reference scan, "NOT what anyone deploys" (both gemmseq boards).
NVIDIA H100, gmm, gp, nystroem, rbfsampler, resample, spectral	RAPIDS ships none of them; the sklearn and scipy CPU arms are refused under GPU-PATH-ONLY (`2026-08-28-NVIDIA-classical.md`).
NVIDIA or AMD, gbm-bench harness (year, covtype, higgs at 500 rounds; lgbm-*-gpu six-arm forest interleave)	`RF_ET_2026-08-22_lightgbm-pairs.md` states the NVIDIA leg was "ready and waiting on a box"; no `gbm_bench_*` result under `bench/results/` carries an NVIDIA or AMD device record. Every gbm-bench number on disk is the M4.
Apple M4, 15 of 21 classical lanes (cd, cholesky, dbscan, hdbscan, holtwinters, ivf, kde, kmeans, knn, kpss, krr, linkage, metrics, ols, pca, svm)	The peer is cuML, cuVS or torch CUDA, none of which runs on Apple; `NO OPPONENT ON THIS BOX` (`2026-08-28-APPLE-classical.md`). The sklearn comparison for ols, pca, kmeans, knn, dbscan is the 2026-08-20 algorithm-matched board, which is a different harness and predates the tier split.
Apple M4, cholesky	torch was not installed in the classical env for that leg (`ModuleNotFoundError("No module named 'torch'")`).
Apple M4, kpss	cuML absent and `statsmodels` not installed.
Apple M4, mamba	`einops` missing took the torch reference arm down (`BOARD_2026-08-28_three-vendor.md` defect 7).
Apple M4, depthwise and lossguide on higgs; any 2M or 5M rung	Only the year fixture has been run for those two lanes on the Mac. "No 5M rung on the Mac by policy, that load goes to rented boxes" (`2026-08-26-APPLE-HIGGS-1M.md`).
Apple M4, rf against lightgbm-cpu with more than one round	The lightgbm-cpu arm ran 258 s per fit and hit the per-arm budget after one round (`2026-08-28-APPLE-forest.md`).
Apple M4, xgboost CPU or GPU, cuML, lightgbm-cuda	Not installed or no Metal build; refused by name on every Apple tree board.
ridge, logistic, arima, svr on any column	No speed lane exists. `ARIMA` has no `fit` and `SVR` is absent (`CHANGELOG.md` 0.2.0).
Any lane, IDENTICAL against a peer other than gemm	The only IDENTICAL-versus-vendor timings on disk are the GEMM shapes in `SPEED_LANE_2026-08-25.md`. Every other lane's IDENTICAL cost is ours-versus-ours (section 4).
Scale label on every forest board	Every 2026-08-28 forest board prints "Scale UNKNOWN for every row in this run" because the arms did not emit `scale=`; the rows are listed as fixed-cost "because that is the conservative bucket", not because they are small.

4. The IDENTICAL price, where it was measured

All of these are ours against ours, same box, same shapes, two binaries, except the SPEED_LANE section 3 rows already listed in 2.18.

4.1 GEMM fold pin, `gemm/mojo_only/gemm_unpinned_price.mojo`

pinned over unpinned at the tiled `llama8b` t512 rows, 2026-08-25 (`SPEED_LANE_2026-08-25.md` 1.2, 1.2a, 1.2c).

shape	Apple M4	NVIDIA H100	AMD MI325X
qkv.t512	1.522x	2.33x	2.21x
mlp_up.t512	1.538x	2.32x	2.20x
mlp_down.t512	1.546x	2.31x	2.17x
lm_head.t512	1.542x	2.32x	2.06x
seams share of the pin (pin over strict, t512)	1.24x of 1.55x	1.83x of 2.31x	2.12x of 2.21x
mlp_down.t1 pin over unpinned	not listed	1.02x	2.30x

The file's own sentences to carry. "The pin costs 2.31x on an H100 against 1.55x on an M4." "AMD sits with NVIDIA, not with Apple." "Report the range with the table, never a scalar." The M4 prediction on record was 1.10x to 1.45x and it was wrong on the low side. Confounds are listed in 1.2b.

4.2 Per-lane price, `bench/results/lanes_price/`

FAST versus IDENTICAL, ours only, Apple M4, 5 rounds, commit 26eb8ba, 2026-08-28 (`lanes_price/2026-08-28_163353_26eb8ba/ratio.tsv`). The M4 drift caveat applies; two modes are two binaries and cannot interleave inside one process.

lane	FAST median s	IDENTICAL median s	ratio (min to max)	bits
gemm (`gram.32x32x1M` , split-K 240 chunks against pinned 128)	0.000710	0.003297	4.644x (3.961 to 5.849)	fast == ident
cd	0.007365	0.009967	1.353x (1.199 to 1.714)	fast != ident
kde	0.003200	0.003799	1.187x (0.984 to 1.343)	fast != ident
svm	0.018298	0.019268	1.053x (0.791 to 1.206)	fast != ident
linkage	0.011320	0.011176	0.987x (0.703 to 1.009)	fast == ident
metrics	0.012427	0.011196	0.901x (0.721 to 1.194)	fast != ident

The 2026-08-23 smoke at e69db89 (1 round, `lanes_price/2026-08-23_125704_e69db89_SMOKE/ratio.tsv`) read gemm 1.545x, cd 1.530x, kde 1.007x, linkage 1.014x, svm 0.944x, metrics 0.788x; the 26eb8ba run is current.

4.3 Two named single-shape prices

`gemm/IDENTICAL_FP32_CONTRACT.md` clause 5 records "the 2.85x pinned-distance cost" (IDENTITY_PATHS row 24, the k-NN distance step) and "the 4.7x measured on `nt.4096x64x64` " (the linalg lane's `gemm_nt` , MAX matmul under FAST against `pinned_gemm_nt_kernel` under IDENTICAL, DEVIATION 526), and says not to present either as universal. `bench/linalg_price_main.mojo` and `bench/gemm_price_main.mojo` carry the same two figures in their docstrings. No dated board under `bench/results/` records either measurement; the driver text places them on the M4.

4.4 What the flag buys, same box, same run

`BOARD_2026-08-28_three-vendor.md` 1.3. gemm versus its oracle on Apple, FAST 31 shapes differ, IDENTICAL 0 differ; on AMD FAST 41 differ, IDENTICAL 0. Transformer clause (d) fails under FAST and passes under IDENTICAL. k-NN on one tie-rich fixture returns three different sorted index sets in three FAST runs and one under IDENTICAL. `stability/RESULTS.md` (2026-08-29) reads FAST moved in 8 of 10 attempts on the M4, 24 different answers in 24 calls on an RTX 4090, 6 answers in 24 on the MI325X; deterministic and identical stable on all three.

5. Anomalies the boards themselves flag

- Extra trees on the MI325X is 4.40x slower than on the H100 at higgs 1M (18294.084 versus 4160.298 ms) and 4.64x at 2M, same source, while rf and iforest are 1.17x and 1.50x faster on AMD. "An AMD-specific ET problem with a number on it for the first time" (`BOARD_2026-08-28_three-vendor.md` 2.3).
- The 5M rung. Lossguide read 14800.158 ms with min 6030.534 at 69f2503 on 2026-08-27 (5.59x behind xgboost-gpu) and 6301.633 ms at 3d65d70 the next morning (2.61x). The symmetric arm bent superlinear between 2M and 5M at 374875a (marginal 273.6 then 1389.6 ms per 1M rows) and straightened to 518 and 525 after 132d754 (`2026-08-26-nvidia-HIGGS-ladder-2M.md`).
- The same cell on the same day, two H100 hosts. gbdt-symmetric higgs 1M read 3666.967 ms (min 2622.429, warm-up 2125.632) at 3d65d70 in the 030908 leg and 856.116 ms at a8d838e in the 040157 leg. The 2M row in the 030908 leg is 1304.338, faster than its own 1M row. The ladder file's finding 1 explains the mechanism, "RunPod hosts differ in CPU allocation and our arm is host-prep-heavy"; the 1.86x-faster crossover at c637e30 became 1.19x slower at 374875a on a different host, so cross-leg comparisons are "for direction only".
- The M4 drifts. "This box has been measured drifting 1.7x inside twenty minutes when heat pins the GPU governor" (`2026-08-26-APPLE-trees.md`); the same binary measured 21.1 then 12.1 ms per tree nineteen minutes apart (`PERF_2026-08-20_fixed-cost.md`). Only ratios from arms alternated inside one process are quoted; the six interleaved year runs on 2026-08-22 ranged 1.14x to 1.68x (`README.md`). The 2026-08-27 covtype rf window is not claimed for the same reason (sklearn 9.1 s to 17.4 s day over day, `2026-08-27-APPLE-covtype-et-flip.md`).
- The H100 IDENTICAL GEMM variance. Two single-round legs an hour apart at the same commit read 4.90 and 8.799 ms at `llama8b.qkv.t512`, a 1.8x spread, so every H100 row in `SPEED_LANE_2026-08-25.md` is a band and the M4 and MI325X rows are single-round; "Re-run before publication".
- The FAST GEMM is TF32 on NVIDIA and was not declared until DEVIATION 1885. MAX `linalg.matmul` measured 200 TFLOP/s against an H100 FP32 peak of 67. The 535x transformer speedup at t1 carries a 47x to 239x worsening of max_abs_diff against the fp32 reference, with rmsnorm (no GEMM) unchanged as the control (`2026-08-26-DEVIATION-1885-fast-gemm-is-tf32.md`).
- `check_device_is_batch_invariant` fails under FAST on NVIDIA and AMD and holds on Apple, "a real defect" and not a FAST wobble; round 13 verdict NOT-CLOSED (`BOARD_2026-08-28_three-vendor.md` 1.4). `check_ols_is_launch_invariant` fails on the H100 and MI325X in both modes (`CHANGELOG.md` known issues).
- k-NN own-arm dispatch on Apple. `ours-tiled` is 1.12x faster than the shipped `ours` on the M4 and its hash moved in 3 of 3 rounds; on the H100 the tiled arm is 2.04x slower than shipped (`2026-08-28-APPLE-classical.md`, `2026-08-28-NVIDIA-classical.md`). The kernel-matrix rows are per vendor.
- The knn H100 row moved 21.6x to 23.9x slower (2026-08-25 and 26) to 2.76x slower (2026-08-28) after DEV 1921-1923; the older files are superseded.
- The cuML k-means arm moved 236.789 ms to 120.217 ms between the 2026-08-26 and 2026-08-28 H100 legs while ours held at 155 ms, which flipped the cell from 1.53x faster to 1.29x slower.
- Fixed-cost rows are unclaimable in both directions. Every 2026-08-28 classical board ranks its small fixtures under "NOT a speed claim"; the Apple six-lane file says the 12x to 29x sklearn wins there are "the same mechanism as our 2-16x wins over cuML-GPU on small fixtures on the H100, with the roles reversed".
- Peers with n=1. lightgbm-cuda in the ET lane (181 s and 227 s per fit) and lightgbm-cpu in the Apple rf lane (258 s) each completed one timed round before the per-arm budget cut them off; the 36.69x, 30.53x and 16.89x ratios stand on a single peer sample.
- Nine silent harness defects on 2026-08-28 (gemm_nt_gram IDENTICAL not compiling since 2026-08-25, `git rev-parse` killing the NVIDIA matrix, iforest cards written to a scratch path, einops, the seq torch arm refusing non-CUDA boxes, the AMD leg with no dataset download) all produced boards "full of NO OPPONENT or plausible rows rather than errors" (`BOARD_2026-08-28_three-vendor.md` part 3).
- 2026-08-25 NVIDIA tree numbers (`2026-08-25-nvidia-trees-PARTIAL.md`, 4cbead3) were transcribed from a live terminal after the leg script died before its fetch; they are not card-backed and every later leg supersedes them. Its CPU rows are struck.
- Accuracy where it is not fine. higgs gbdt AUC 0.822 versus CatBoost 0.830 on the gbm-bench harness is an open parity gap (`README.md`, `THREE_SUITE_QUIET_2026-08-22.md`); year ET on the LightGBM pairs suite is behind on MSE (98.0 versus 92.8). Every 2026-08-28 higgs row on the H100 is at parity or better on logloss and AUC per the boards' accuracy tables.

6. Source files read

- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/BOARD_2026-08-28_three-vendor.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/SPEED_LANE_2026-08-25.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/BOARD_2026-08-20_algorithm-matched.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/SCOREBOARD_2026-08-19.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/SKLEARN_GPU_BASELINE_2026-08-20.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/SCALING_2026-08-19.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/SCALING_2026-08-20_catboost.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/DBSCAN_RBC_2026-08-19.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/GEMM_ROUND_2026-08-19.md`

- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/PERF_2026-08-20_fixed-cost.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/PERF_2026-08-20_estimation-bill.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/WINDOW_2026-08-20_interleaved-200k.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/COVTYPE_MULTICLASS_2026-08-21.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/EPSILON_2026-08-21_interleaved.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/EPSILON_DEPTH8_2026-08-21.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/EPSILON_LOGLOSS_2026-08-21.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/SHAPE_SWEEP_2026-08-21_epsilon.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/EXTRATREES_SKLEARN_2026-08-21.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/RF_2026-08-21_pipelined-forest.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/RF_2026-08-22_attribution.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/RF_ET_2026-08-22_lightgbm-pairs.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/WINDOW_2026-08-22_extratrees-batched.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/GBM_BENCH_2026-08-22_year.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/THREE_SUITE_2026-08-22.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/THREE_SUITE_QUIET_2026-08-22.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/LANE_knn-speed-campaign_2026-08-28.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-25-nvidia-trees-PARTIAL.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-25_155542-nvidia-gemmseq.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-25_160520-nvidia-gemmseq.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-25_235543-nvidia-classical.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26-APPLE-HIGGS-1M.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26-APPLE-trees.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26-DEVIATION-1885-fast-gemm-is-tf32.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26-nvidia-HIGGS-ladder.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26-nvidia-HIGGS-ladder-2M.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26-nvidia-forest-postfix.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26_040049-nvidia-trees-ladder.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26_040054-nvidia-gemmseq.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26_040100-nvidia-classical.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-26_162556-nvidia-forest.md` (via its summary file `2026-08-26-nvidia-HIGGS-ladder-2M.md`)
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-27-APPLE-classical-six-small-lanes.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-27-APPLE-covtype-et-flip.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-27-APPLE-trees-evening.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-27_031904-nvidia-forest.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-27_171939-nvidia-depthwise.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-AMD-forest.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-AMD-forest-higgs.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-APPLE-classical.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-APPLE-forest.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-APPLE-gemmseq.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-NVIDIA-classical.md`
- `/Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-NVIDIA-forest-A.md`

- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-NVIDIA-forest-B.md
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28-NVIDIA-gemmseq.md
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28_030908-nvidia-forest.md
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/2026-08-28_030911-nvidia-forest.md
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/mac-higgs-2026-08-26/table.md
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/do-2026-08-28_125022-amd-forest/run/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/do-2026-08-28_130709-amd-forest/run/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/do-2026-08-28_140119-amd-forest/run/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/mac-2026-08-28_040202-forest/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/mac-2026-08-28_122551-classical/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/mac-2026-08-28_122711-gemmseq/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/fast_speed/mac-2026-08-28_131915-gemmseq/leg.txt
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/e1g/2026-08-2*-nvidia-speed-*/leg.txt (the ten NVIDIA legs, for commit and round provenance)
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/gemm_ladder/2026-08-23_095619_fe00e8a-dirty/ladder.card
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/gemm_ladder/2026-08-23_101159_ec944d8-dirty/ladder.card
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/gemm_ladder/2026-08-23_101329_5c6932b-dirty/clean.card and sab.card (identity cards, no timings)
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/lanes_price/2026-08-23_125704_e69db89_SMOKE/env.txt, summary.tsv, ratio.tsv
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/lanes_price/2026-08-28_163353_26eb8ba/env.txt, summary.tsv, ratio.tsv
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/results/stability/RESULTS.md
- /Users/andrewhendel/CascadeProjects/mojolearn/README.md (Hardware, Benchmarks, Limitations sections)
- /Users/andrewhendel/CascadeProjects/mojolearn/CHANGELOG.md (0.2.0 cross-vendor standing, third tier, Performance, Known issues)
- /Users/andrewhendel/CascadeProjects/mojolearn/gemm/IDENTICAL_FP32_CONTRACT.md (clause 5, for the 2.85x and 4.7x figures)
- /Users/andrewhendel/CascadeProjects/mojolearn/bench/linalg_price_main.mojo and bench/gemm_price_main.mojo (docstrings, same two figures)

Skimmed for ours-versus-peer cells and found none. The twenty LANE_*_2026-08-19 and LANE_*_2026-08-20 files (ball-cover, covariance, dbscan, decomposition, dispatch-audit, gram, kmeans, knn, pca, rbc, splitk, target-keying, vendor-correctness, warpsort) are Apple work logs whose only ratios are ours-against-ours kernel A/Bs (for example the ball-cover launch shape sweep, "1.54x slower at n = 16,000" for their tpb=64 against one query per block, LANE_rbc-build_2026-08-19.md); their peer comparisons were folded into SCOREBOARD_2026-08-19.md and the 2026-08-20 algorithm-matched board. The WINDOW_2026-08-20_*, FIRST_RUN, FUSED_ROUND, SUBSTITUTION_ROUND, VENDOR_PATH, GRAM_PROFILE, MODULAR_UPSTREAM, PREP_BILL, CONTAMINATION, DEFECT, RF_2026-08-21_accuracy-and-profile and RF_2026-08-21_control-plane files are pre-tier Apple windows whose cells all have a newer source in the tables above.